NETWORK GEO LOCATION ANALYSIS USING WIRESHARK PACKET CAPTURE

A Main project Submitted

By

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Bachelor of Computer Applications

By

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DECLARATION

I hereby declare that the project entitled **"Network Geo Location Analysis Using Wireshark Packet Capture"** is an original work carried out by me as part of the Cybersecurity course requirements. This project has not been submitted previously, either in part or in full, to any other institution or university for the award of any degree, diploma, or certification. All sources of information used in this project have been properly acknowledged in the references section.

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CERTIFICATE

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Abstract

Cybersecurity has become one of the most critical aspects of modern networked systems due to the rapid increase in cyber threats and malicious activities. Among various types of network-based attacks, TCP SYN flooding is a common Denial of Service (DoS) technique that targets system availability by overwhelming a victim machine with excessive connection requests. This project focuses on detecting TCP SYN flooding attacks using Wireshark packet capture and performing network geolocation analysis to identify the origin of suspicious traffic. In this study, a controlled TCP SYN flood attack was simulated in a secure laboratory environment using Kali Linux and the `hping3` tool. The attack was carefully executed to ensure that it remained within a private network and did not affect external systems. During the execution phase, Wireshark was used on the victim machine to capture real-time network traffic in PCAP format. The captured packet data was then analyzed using appropriate display filters to isolate TCP SYN packets and identify abnormal traffic behavior. Through detailed packet inspection, key information such as source IP address, destination IP address, protocol type, and TCP flags were extracted. Additionally, the Ethernet frame headers were examined to obtain the attacker's MAC address within the local network. The unusually high number of SYN packets transmitted within a short time interval confirmed the presence of flooding behavior. This analysis demonstrates how packet-level monitoring can be used to detect and understand network-based attacks effectively. Furthermore, network geolocation analysis was performed using GeoIP techniques to determine the approximate geographical location of the attacker's IP address. By integrating IP address data with geolocation databases, the project visualized the possible origin of the network traffic. Although geolocation results are limited to ISP-level approximation and cannot reveal an exact physical address, they provide valuable insight into the regional source of suspicious activity. Overall, this project highlights the importance of packet capture, traffic analysis, and geolocation mapping in modern cybersecurity practices. The study demonstrates how tools such as Wireshark can be used not only to detect attacks but also to support basic digital forensics and network investigation processes in a controlled and ethical manner.

1. Introduction

A SYN flood attack is a type of Denial of Service (DoS) attack that exploits the TCP three-way handshake process used to establish reliable network communication. In normal TCP communication, a connection between a client and a server is initiated through a three-step process: the client sends a SYN (synchronize) packet, the server responds with a SYN-ACK (synchronize-acknowledgment) packet, and finally, the client replies with an ACK (acknowledgment) packet to complete the connection. This mechanism ensures reliable and ordered data transmission across networks.

However, in a SYN flood attack, the attacker deliberately sends a large number of SYN packets to the target system but does not complete the handshake process. Instead of responding with the final ACK packet, the attacker either ignores the server's SYN-ACK response or spoofs the source IP address. As a result, the server keeps multiple half-open connections in its memory, waiting for acknowledgments that never arrive. Over time, these incomplete connections consume system resources such as memory buffers and processing power. When the number of half-open connections increases significantly, the victim system may become slow, unresponsive, or completely unavailable to legitimate users. This leads to denial of service, where authorized users are unable to access network services. SYN flooding is particularly dangerous because it can be launched with relatively low resources while causing significant disruption to the target system. With the increasing dependency on online services, understanding SYN flood attacks has become essential in cybersecurity. Organizations must be capable of detecting abnormal traffic patterns and responding effectively to such threats. Network packet analysis tools like Wireshark play a crucial role in identifying flooding behavior by capturing and analyzing real-time traffic data.

In this project, a controlled SYN flood attack is simulated within a secure laboratory environment to study its behavior and detection mechanisms. By analyzing packet-level details such as TCP flags, source IP addresses, and packet transmission rates, it becomes possible to recognize flooding characteristics. This foundational understanding is important for developing intrusion detection strategies and strengthening network security measures against denial-of-service attacks.

2. Tools Used

This project utilizes several specialized tools and technologies to simulate, capture, and analyze network traffic during a TCP SYN flooding attack. Each tool plays a critical role in the successful execution and investigation of the attack scenario.

2.1 Kali Linux

Kali Linux is a widely used penetration testing and cybersecurity-focused operating system based on Debian Linux. It contains numerous pre-installed tools designed for network security assessment, vulnerability analysis, and ethical hacking. In this project, Kali Linux was used as the attacker machine to simulate the TCP SYN flooding attack. Its built-in networking utilities and security testing environment provide a controlled and flexible platform for conducting attack simulations in a safe laboratory setup.

2.2 hping3

hping3 is a command-line network tool used for crafting and sending custom TCP/IP packets. It is commonly used for network testing, firewall evaluation, and security auditing. In this project, hping3 was used to generate a high-volume TCP SYN flood attack. By using the `-flood` option along with the SYN flag, the tool transmitted a large number of SYN packets to the target system at maximum speed. This allowed the simulation of resource exhaustion and abnormal traffic patterns required for detection analysis.

2.3 Wireshark

Wireshark is a powerful open-source network protocol analyzer used for capturing and examining real-time packet data. It allows users to inspect network traffic at a very detailed level, including protocol headers, IP addresses, TCP flags, and packet timing information. In this project, Wireshark was installed on the victim machine to capture the incoming flood traffic.

Display filters were applied to isolate TCP SYN packets, and detailed packet inspection was performed to identify suspicious patterns. Wireshark also provided endpoint statistics and geolocation integration features for further analysis.

2.4 GeoLite2 Database

The GeoLite2 database, developed by MaxMind, is a free IP geolocation database that maps IP addresses to approximate geographic locations. It provides country, region, city, and ISP-level information based on public IP addresses. In this project, the GeoLite2 database was used to perform network geolocation analysis of the attacker's IP address. Although geolocation data provides only approximate location information, it helps in visualizing the potential origin of suspicious network traffic.

2.5 Local Network Setup

A controlled local network environment was established to safely simulate and analyze the attack. The attacker system (Kali Linux) and victim system were connected within the same private network. This ensured that the attack remained isolated from external networks and did not impact real-world systems. Conducting the experiment in a local setup allowed safe testing, packet capture, and analysis without violating legal or ethical guidelines.

3. Attack Execution

In order to study and analyze TCP SYN flooding behavior, a controlled attack was simulated within a secure local network environment. The attack was executed from the Kali Linux attacker machine using the `hping3` network testing tool. The following command was used to generate the flooding traffic:

```
sudo hping3 -S --flood -p 80 192.168.127.128
```

3.1 Command Explanation

Each parameter in the command plays a specific role in generating the flooding attack:

- `sudo`: Executes the command with administrative privileges, which are required for crafting and sending raw network packets.
- `hping3`: The packet crafting tool used to generate customized TCP/IP packets.
- `-S`: Sets the TCP SYN flag, initiating connection requests similar to the first step of the TCP handshake.
- `-flood`: Sends packets as fast as possible without waiting for responses, creating high-volume traffic.
- `-p 80`: Targets port 80 on the victim machine, which is commonly associated with HTTP services.
- `192.168.127.128`: The IP address of the victim system within the local network.

3.2 Attack Behavior

When the command was executed, hping3 continuously transmitted TCP SYN packets to the target system at maximum speed. Because the `-flood` option was enabled, the tool did not wait for responses from the victim machine. Instead, it generated a large volume of connection requests in a very short time period.

The output statistics showed that millions of packets were transmitted during the attack session. The absence of received packets and the display of 100% packet loss is expected behavior in flood mode, as the tool prioritizes packet transmission speed over response handling.

3.3 Impact on Victim System

The victim machine received an unusually high number of SYN packets within a short interval. Since each SYN packet initiates a half-open TCP connection, the system temporarily allocates resources while waiting for the handshake to complete. When the handshake is not completed, these half-open connections accumulate, potentially leading to resource exhaustion.

Although this experiment was conducted in a controlled laboratory setup, it successfully demonstrated the fundamental mechanism behind SYN flooding attacks. The generated traffic provided sufficient data for packet capture and analysis using Wireshark in subsequent stages of the project.

3.4 Ethical Considerations

It is important to note that the attack simulation was performed strictly within a private network environment. No external systems were targeted, and the experiment was conducted solely for educational and research purposes. Ethical guidelines were followed to ensure that the testing process remained safe and legally compliant.

Terminal Output

```
HPING 192.168.127.128 (eth0 192.168.127.128): S set, 40 headers + 0 data
bytes
hping in flood mode, no replies will be shown
70512387 packets transmitted, 0 packets received, 100
```

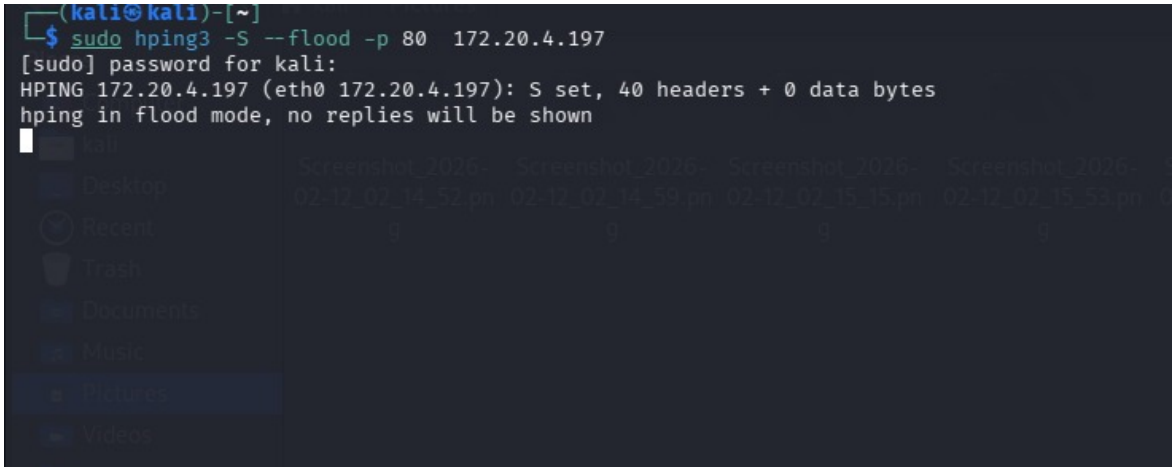

Figure 3.1: TCP SYN Flood Attack Execution in Kali Linux

4. Packet Capture and Filtering

In order to analyze the network traffic generated during the TCP SYN flooding attack, Wireshark was installed and executed on the victim system. Wireshark is a powerful network protocol analyzer that captures real-time packet data and allows detailed inspection of network communication at various protocol layers.

4.1 Packet Capture Process

Before initiating the attack, Wireshark was started on the victim machine to ensure that all incoming and outgoing traffic would be recorded. The appropriate network interface (Ethernet or Wi-Fi) was selected based on the active connection. Once the interface was selected, packet capturing was initiated.

During the attack execution phase, Wireshark continuously recorded all network packets transmitted to and from the victim system. The captured data included detailed information such as source and destination IP addresses, protocol types, TCP flags, sequence numbers, and timestamps. This raw packet data was stored in PCAP format for further analysis.

4.2 Display Filtering

Because network traffic can contain thousands of different packets from various protocols, filtering is necessary to isolate relevant traffic. In this experiment, the focus was specifically on identifying TCP SYN packets associated with flooding behavior.

The following display filter was applied in Wireshark:

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

This filter isolates packets where:

- The SYN flag is set (`tcp.flags.syn == 1`)
- The ACK flag is not set (`tcp.flags.ack == 0`)

Such packets represent the initial step of the TCP handshake process. In a SYN flood attack, a large number of these packets are transmitted without completing the handshake, making this filter ideal for detecting flooding behavior.

4.3 Observations

After applying the display filter, a significant number of SYN packets were observed within a very short time interval. The repeated appearance of packets from the same source IP address, along with high transmission frequency, confirmed abnormal traffic patterns.

The packet timestamps indicated extremely small intervals between consecutive SYN packets, demonstrating high-speed transmission characteristic of flooding attacks. Additionally, the lack of corresponding ACK responses suggested incomplete TCP handshakes, further validating the presence of SYN flood behavior.

4.4 Significance of Filtering

Packet filtering plays a critical role in network forensic analysis. Without filtering, identifying suspicious packets within large volumes of normal traffic would be difficult and time-consuming. By isolating SYN packets, the analysis became more focused and efficient, enabling accurate detection of attack patterns.

The successful capture and filtering of SYN packets provided essential evidence for subsequent packet inspection, endpoint analysis, and geolocation mapping stages of the project.

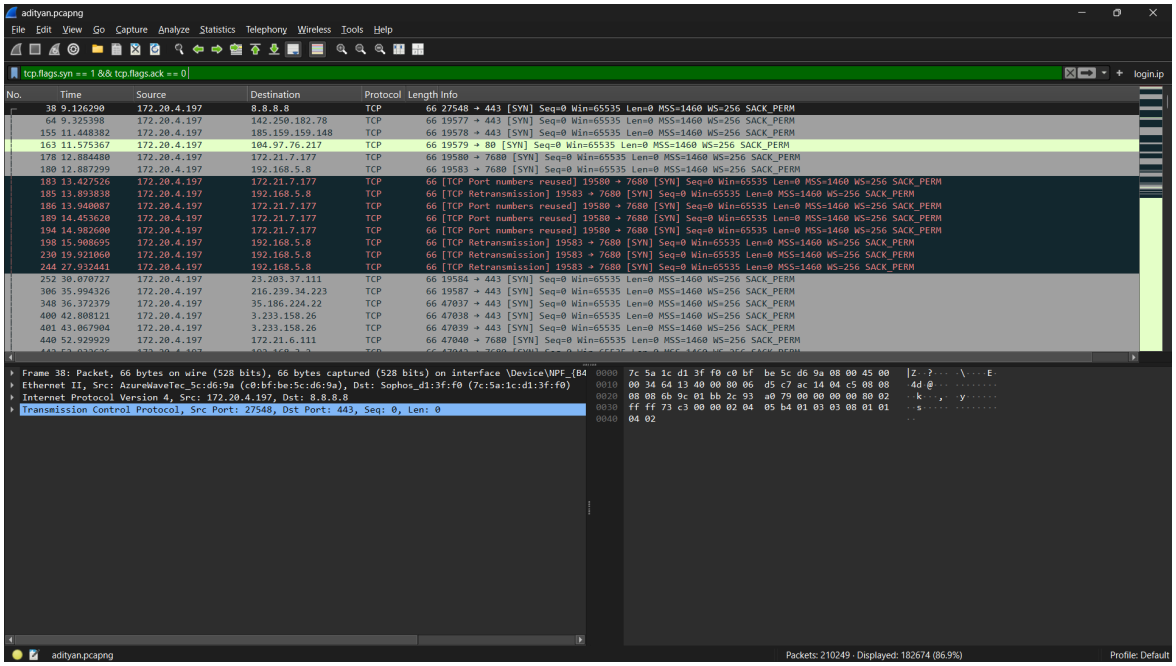

Figure 4.1: SYN Flood Packets Captured in Wireshark

5. Packet Inspection

After successfully capturing and filtering the relevant SYN packets, detailed packet inspection was performed to extract critical information about the attack source. Wireshark provides a layered breakdown of each captured packet, allowing examination of protocol headers at multiple levels. In this phase, the Ethernet II, IPv4, and TCP headers were expanded and analyzed to obtain the source IP address, MAC address, and TCP flag information.

5.1 Ethernet II Header Analysis

The Ethernet II header represents the data link layer (Layer 2) of the OSI model. This header contains the physical (MAC) addresses of both the source and destination devices within the local network.

By expanding the Ethernet II section in Wireshark, the following details were observed:

- Source MAC Address
- Destination MAC Address
- EtherType (indicating IPv4 protocol)

The source MAC address corresponds to the physical network interface of the attacker machine within the local network. Identifying the MAC address is particularly useful in LAN-based investigations, as it helps trace the device responsible for generating the traffic. However, it is important to note that MAC addresses are not visible beyond the local network due to router-based address replacement.

5.2 IPv4 Header Analysis

The IPv4 header operates at the network layer (Layer 3) and provides logical addressing information. Expanding this section revealed:

- Source IP Address
- Destination IP Address
- Time To Live (TTL)
- Protocol Type (TCP)

The source IP address identifies the origin of the SYN packets, while the destination IP corresponds to the victim system. The repeated appearance of the same source IP across thousands of packets confirmed that the traffic was generated from a single attacking system. The TTL value and protocol information further validated that the packets were legitimate TCP packets crafted by the hping3 tool.

5.3 TCP Header Analysis

The TCP header provides transport-layer details related to connection management. Expanding the TCP section revealed important fields such as:

- Source Port Number
- Destination Port Number
- Sequence Number
- TCP Flags (SYN, ACK, etc.)

In this experiment, the SYN flag was consistently set, while the ACK flag remained unset. This confirmed that the packets represented the initial stage of the TCP handshake process. The absence of corresponding ACK responses indicated incomplete connections, which is a defining characteristic of SYN flood attacks.

5.4 Observations and Interpretation

The combination of Ethernet II, IPv4, and TCP header analysis provided comprehensive insight into the attack behavior. The source IP and MAC address were successfully extracted, confirming the identity of the attacking device within the local network. The repeated SYN flag pattern and high packet frequency further validated the flooding mechanism.

Packet inspection is a critical step in network forensics, as it allows investigators to move beyond simple detection and examine the technical structure of malicious traffic. By analyzing protocol headers in detail, it becomes possible to understand how the attack was generated and how it impacts the victim system.

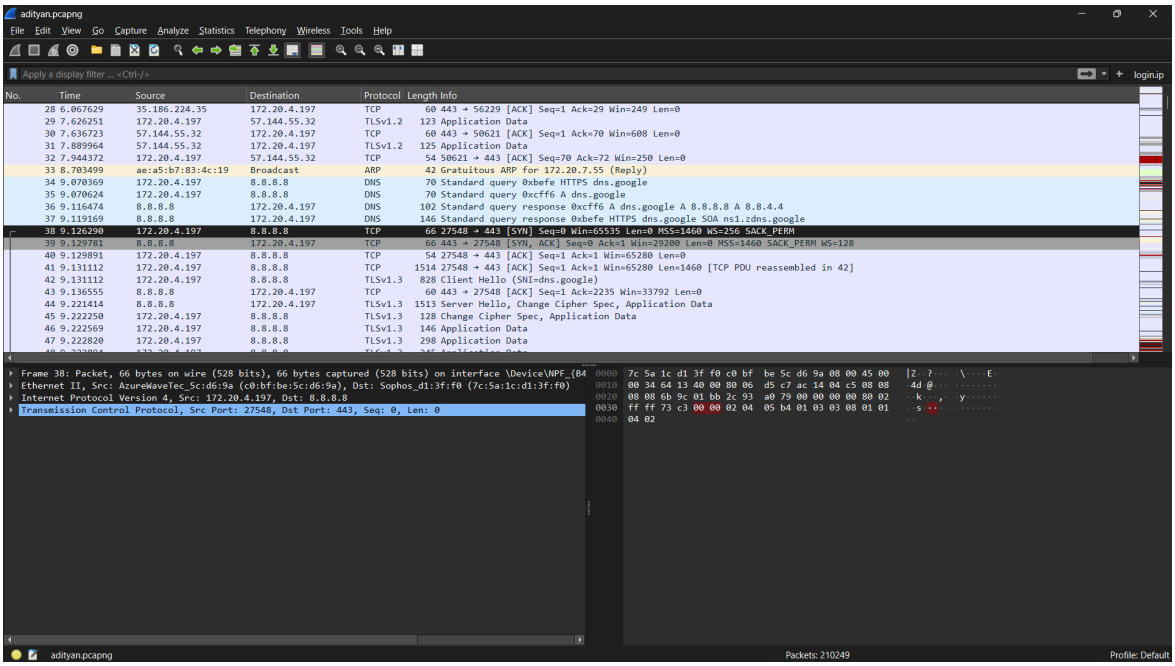

Figure 5.1: Detailed Packet Header Inspection

6. Endpoint Analysis

After capturing and filtering the relevant SYN packets, endpoint analysis was performed to obtain a summarized overview of network communication. Wireshark provides a built-in statistical feature that allows investigators to examine communication patterns between different network entities in a structured manner.

To access the endpoint statistics, the following navigation path was used:

`Statistics → Endpoints → IPv4`

This feature displays a table containing all IPv4 addresses detected in the captured packet data. It provides statistical information about each endpoint involved in the communication process.

6.1 Endpoint Statistics Overview

The IPv4 endpoint table includes important parameters such as:

- IP Address
- Number of Packets Sent and Received
- Total Bytes Transmitted
- Country and City (when GeoIP is enabled)
- Latitude and Longitude
- Autonomous System Number (ASN)

By analyzing the packet count column, the attacking IP address was identified as the endpoint responsible for generating the highest volume of traffic. The unusually large number of transmitted packets compared to other endpoints confirmed abnormal behavior consistent with a SYN flood attack.

6.2 Traffic Pattern Interpretation

The endpoint statistics clearly showed a traffic imbalance between the attacker and victim systems. The attacker transmitted a significantly higher number of packets, while the victim system showed minimal response activity. This asymmetrical traffic pattern is characteristic of SYN flooding, where connection requests are continuously sent without completing the TCP handshake.

Endpoint analysis provided statistical confirmation of the flooding behavior observed during packet filtering and inspection stages.

6.3 Geolocation Integration

When the GeoLite2 database was configured within Wireshark, the endpoint table automatically displayed approximate geographical information for public IP addresses. This allowed the identification of the regional origin of suspicious traffic.

Although geolocation data provides only ISP-level approximation and cannot reveal an exact physical address, it enhances situational awareness and supports network forensic investigation.

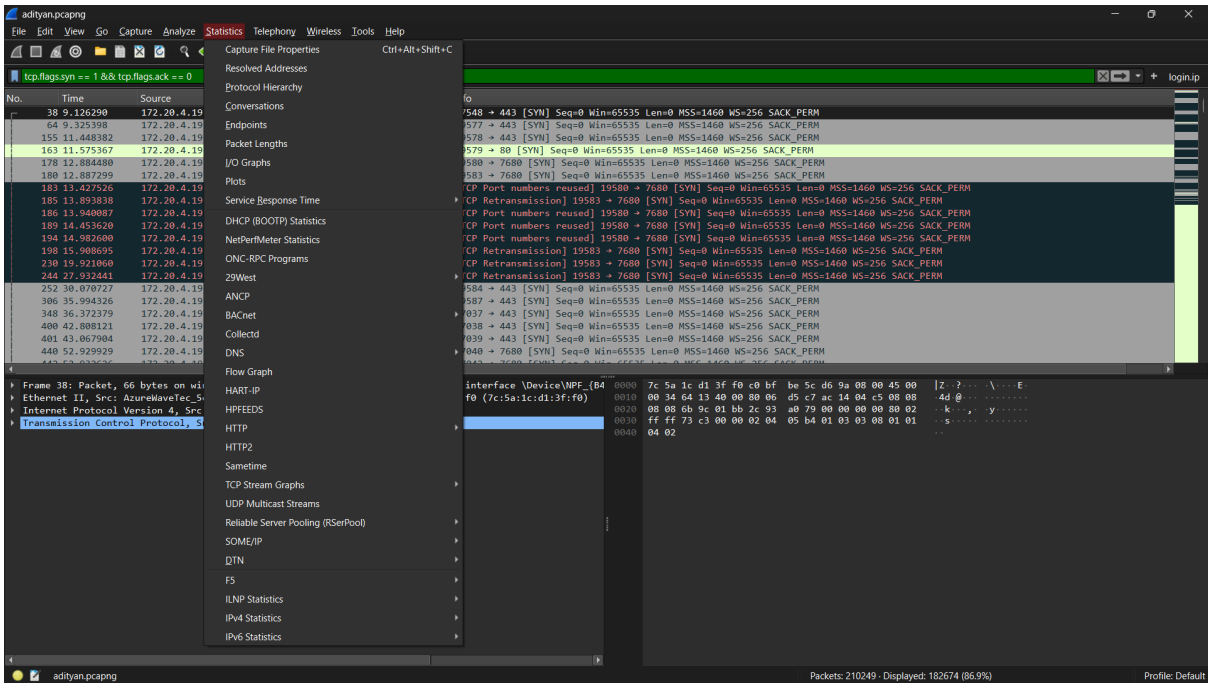

Figure 6.1: Endpoint Statistics with IP Geolocation Data

The above figure illustrates the IPv4 endpoint statistics table showing packet counts and geolocation details associated with the detected IP addresses.

7. Geolocation Mapping

After identifying the attacker's IP address through packet inspection and endpoint analysis, the next step involved determining the approximate geographical origin of the suspicious traffic. This was achieved using IP geolocation techniques integrated with the GeoLite2 database.

7.1 GeoIP Database Integration

The GeoLite2 database, developed by MaxMind, was configured within Wireshark to enable IP-to-location mapping. This database maps public IP addresses to approximate geographic locations based on Internet Service Provider (ISP) allocation records. Once integrated, Wireshark automatically associated captured IP addresses with available location data.

The geolocation information included:

- Country
- Region or State
- City (if available)
- Latitude and Longitude coordinates
- Autonomous System Number (ASN)

7.2 Mapping Process

After enabling GeoIP functionality, the endpoint statistics were revisited. Public IP addresses displayed corresponding geographic data directly within the IPv4 endpoint table. Additionally, Wireshark provides a mapping feature that allows visualization of IP locations on an interactive map.

By selecting the relevant IP addresses and accessing the mapping option, the geographic distribution of network endpoints was displayed visually. This graphical repre-

sentation helped in understanding the regional origin of the traffic involved in the flooding attack.

7.3 Interpretation of Results

The geolocation mapping provided an approximate regional identification of the attacker's IP address. The coordinates and city-level data were based on ISP registration records rather than exact device location. It is important to note that IP geolocation does not reveal precise physical addresses but instead provides an estimation of the general area from which the traffic originated.

In cases where private IP addresses are used within a local network, public geolocation is not possible. Private IP ranges (such as 192.168.x.x, 10.x.x.x, or 172.16.x.x) are reserved for internal network communication and cannot be mapped to public geographic locations. Therefore, geolocation results are only applicable to public IP addresses.

7.4 Significance of Geolocation in Cybersecurity

Geolocation mapping enhances cybersecurity investigations by providing contextual information about suspicious network traffic. While it does not replace advanced forensic investigation, it supports threat analysis by identifying potential regions of origin. This can assist in recognizing unusual traffic sources, detecting coordinated attacks, and improving situational awareness.

In this project, geolocation mapping complemented packet analysis and endpoint statistics, providing a broader understanding of the attack origin. The integration of GeoIP data with packet capture analysis demonstrates how network monitoring tools can be extended beyond detection to include investigative insights.

Overall, geolocation mapping plays a valuable role in modern cybersecurity practices by combining network traffic analysis with geographic visualization, enabling more informed response strategies.

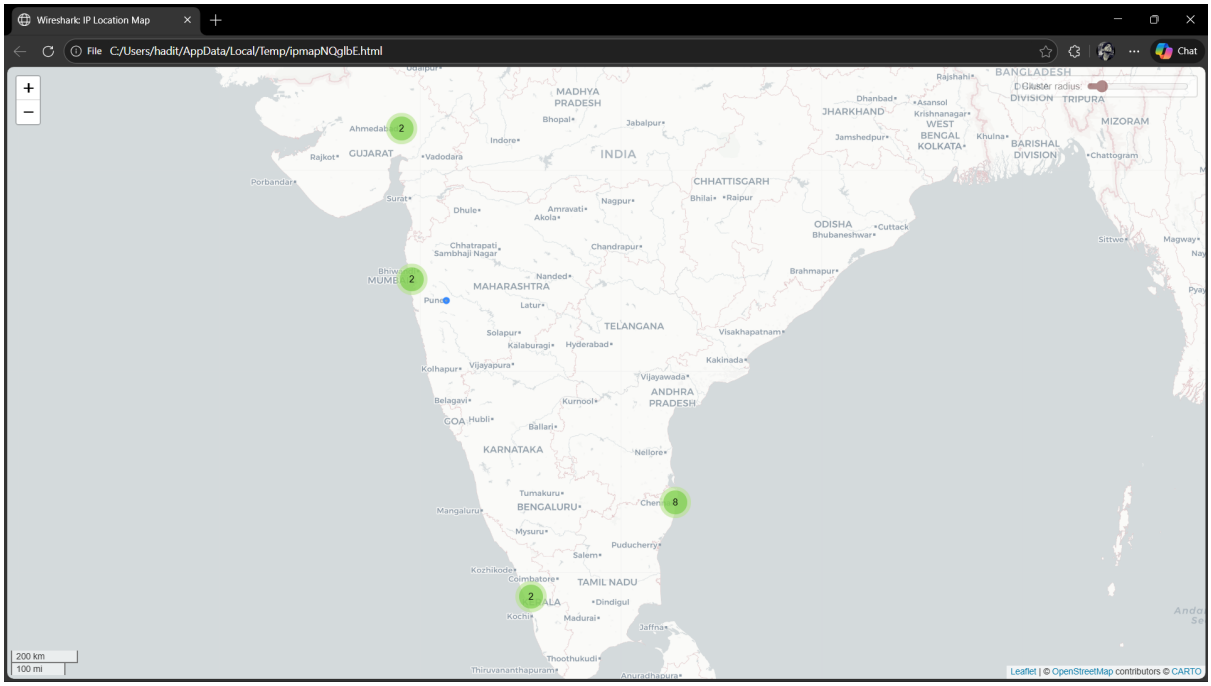

Figure 7.1: Geolocation Map Showing IP Distribution

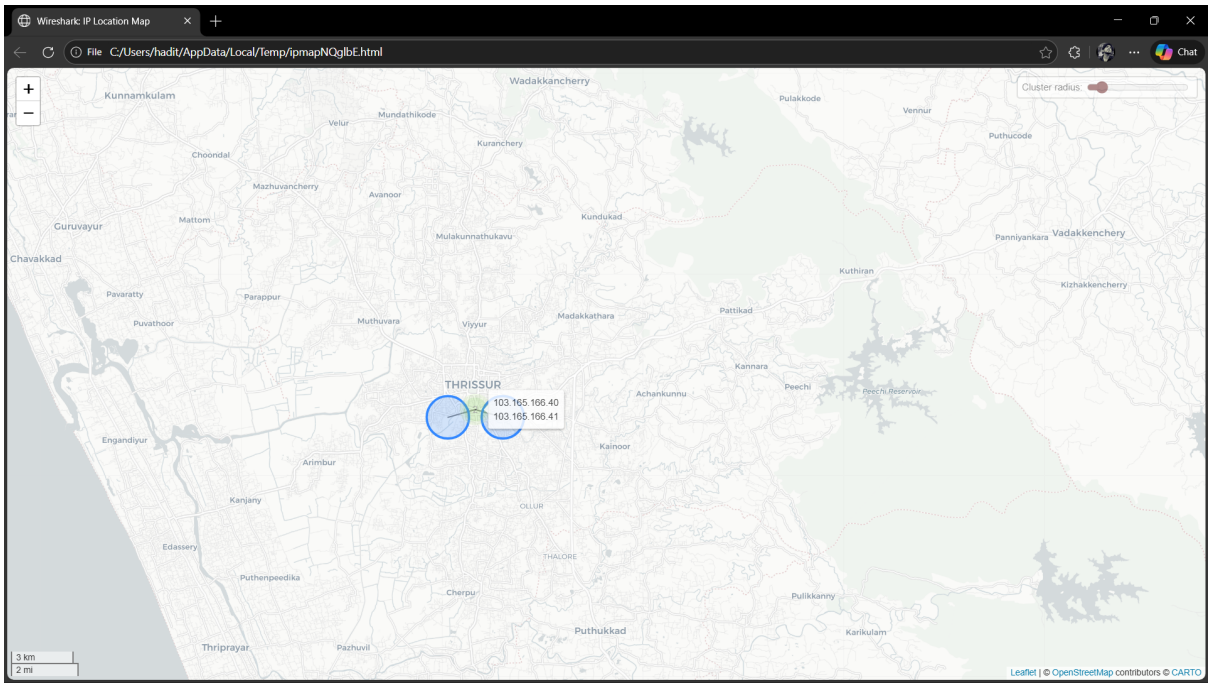

Figure 7.2: Zoomed View of Approximate Location

8. System Architecture and Design

8.1 Overview

The system architecture illustrates the interaction between the attacker system, victim machine, packet capture tool, and geolocation analysis components. The architecture is designed in a controlled laboratory environment to simulate a TCP SYN flood attack and analyze network behavior.

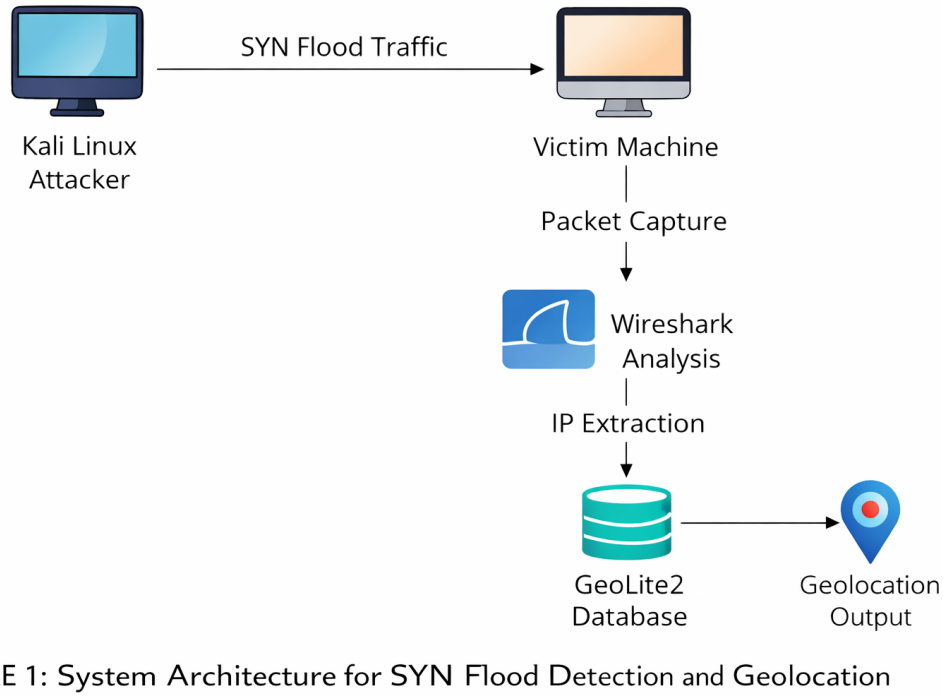

Figure 8.1: System Architecture for SYN Flood Detection and Geolocation Analysis

8.2 Network Setup Verification

The architecture diagram clearly illustrates the controlled laboratory environment used for experimentation. The attacker machine (Kali Linux) and the victim machine were connected within the same isolated private LAN. No external internet-facing systems were involved during the simulation.

This controlled architecture ensures reproducibility of the experiment while maintaining ethical and legal compliance. The diagram provides a visual confirmation of the network topology and traffic flow used throughout the analysis.

8.3 Architectural Components

The system consists of the following primary components:

- Attacker Machine (Kali Linux)
- Victim Machine (Windows 10)
- Wireshark Packet Analyzer
- GeoLite2 Geolocation Database
- Local Network Infrastructure

8.4 System Workflow

The workflow of the system is as follows:

1. SYN flood traffic is generated using hping3 from the attacker system.
2. The victim machine receives high-volume SYN packets.
3. Wireshark captures the incoming traffic.
4. Display filters isolate SYN packets.
5. Endpoint statistics identify attacker IP.
6. GeoIP database maps the IP to an approximate geographical location.

9. Testing and Verification

9.1 Attack Simulation Verification

To verify the successful execution of the TCP SYN flood attack, the packet transmission statistics from the attacker system were observed. The hping3 output confirmed that over 70 million SYN packets were transmitted within a short duration, indicating high-volume traffic generation.

9.2 Packet Capture Verification

Wireshark was used on the victim machine to capture incoming network traffic. The following verification steps were performed:

- Capture initiated before attack execution
- Correct network interface selected
- TCP SYN display filter applied:

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

The large number of SYN packets visible in the capture confirmed attack activity.

9.3 Header Inspection Validation

Ethernet II, IPv4, and TCP headers were expanded to confirm:

- Source IP address consistency
- MAC address extraction
- SYN flag presence
- Absence of ACK completion

This validated that the observed traffic matched SYN flood characteristics.

9.4 Endpoint Statistics Verification

Using:

`Statistics → Endpoints → IPv4`

The attacking IP address showed an unusually high packet transmission count compared to other endpoints, confirming traffic imbalance.

9.5 Geolocation Validation

The GeoLite2 database was used to map the attacker's IP address. The returned country, region, and ASN information were verified against the database output to ensure mapping consistency.

9.6 Result Consistency

All analysis stages — packet capture, filtering, inspection, endpoint statistics, and geolocation mapping — consistently indicated SYN flooding behavior originating from a single source IP within the controlled network environment.

10. Key Findings

The experimental analysis conducted in this project produced several important findings related to SYN flood detection and network geolocation investigation. The following observations summarize the major outcomes of the study:

- SYN flood attack successfully simulated
- Over 70 million packets transmitted
- Attacker IP identified
- MAC address extracted
- Approximate geographical location mapped

10.1 Successful Simulation of SYN Flood Attack

The controlled execution of the TCP SYN flood attack using the `hping3` tool demonstrated that high-volume connection requests can be generated efficiently within a short time interval. The attack simulation remained confined to the local laboratory environment, ensuring ethical compliance while effectively reproducing real-world flooding behavior.

10.2 High Packet Transmission Volume

During the execution phase, the attacker system transmitted over 70 million TCP SYN packets. This extremely high packet count clearly demonstrated the impact of the `-flood` option, which sends packets at maximum speed without waiting for responses. The large transmission volume created measurable abnormal traffic patterns that were successfully captured and analyzed using Wireshark.

10.3 Identification of Attacker IP Address

Through packet inspection and endpoint analysis, the source IP address responsible for generating the flooding traffic was identified. The repeated occurrence of this IP address across millions of packets confirmed it as the origin of the attack. This step is crucial in network forensic investigations, as identifying the source of suspicious traffic is the first stage of threat attribution.

10.4 Extraction of MAC Address

By expanding the Ethernet II header in Wireshark, the physical (MAC) address of the attacking device within the local network was extracted. MAC address identification is particularly useful in local network environments, where devices can be traced at the data-link layer. However, it is important to note that MAC addresses are not visible beyond the local network due to routing mechanisms.

10.5 Geographical Location Mapping

Using the GeoLite2 database, the attacker's IP address was mapped to an approximate geographic location. The mapping provided ISP-level country and regional information, along with latitude and longitude coordinates where available. Although geolocation data does not reveal exact physical addresses, it enhances situational awareness and supports cybersecurity analysis by identifying the potential regional origin of suspicious traffic.

10.6 Overall Impact of Findings

The combination of packet capture, statistical endpoint analysis, header inspection, and geolocation mapping provided a comprehensive understanding of the SYN flood attack. The findings demonstrate that network packet analysis tools such as Wireshark can effectively detect abnormal traffic behavior and assist in identifying potential attack sources. These results highlight the importance of proactive network monitoring and structured forensic analysis in modern cybersecurity environments.

11. Limitations

Although the project successfully demonstrated SYN flood detection and IP geolocation analysis, there are several limitations that must be considered when interpreting the results.

- **Exact physical address cannot be determined**
  IP geolocation techniques do not provide an exact physical address of the attacker. The mapping is based on ISP registration data and network allocation records. Therefore, the identified location represents an approximate geographic area rather than a precise street-level address. Advanced investigative procedures and legal authority would be required to trace an exact physical device.

- **Geolocation is ISP-based and approximate**
  The GeoLite2 database and similar GeoIP services rely on publicly available ISP allocation data. These databases map IP addresses to regions where Internet Service Providers are registered, not necessarily where the device is physically located. As a result, the displayed location may indicate the ISP's operational center rather than the true location of the attacker.

- **Private IP addresses cannot be publicly located**
  Private IP addresses such as 192.168.x.x, 10.x.x.x, and 172.16.x.x are used only within internal networks and are not routable over the public internet. Because of this, they cannot be mapped using public geolocation databases. In local network environments, only MAC address tracking and network-level investigation can identify the specific device.

- **Possibility of IP spoofing**
  Attackers may use IP spoofing techniques to falsify the source IP address of packets. In such cases, the IP address observed in Wireshark may not represent the actual origin of the attack. This makes attribution more complex and may require advanced traffic analysis or intrusion detection systems.

- **Use of VPNs or Proxy Servers**
  An attacker may use Virtual Private Networks (VPNs), proxy servers, or anonymization services. If this occurs, the geolocation result will reflect the location of the VPN or proxy server rather than the true physical location of the attacker.

- **Controlled Laboratory Environment**
  The experiment was conducted within a private and controlled local network. While this ensures safety and ethical compliance, it may not fully represent real-world attack scenarios where multiple distributed sources or large-scale botnets are involved.

- **Resource and Hardware Constraints**
  The testing environment was limited to available system hardware and network capacity. In real-world scenarios, SYN flood attacks can involve significantly higher traffic volumes and distributed attack sources, which may produce different behavioral patterns.

- **Dependency on Database Accuracy**
  The accuracy of geolocation results depends heavily on the reliability and regular updates of the GeoLite2 database. If the database is outdated or incomplete, the mapped location information may be inaccurate or misleading.

- **Lack of Real-Time Intrusion Prevention**
  The project primarily focused on detection and analysis rather than prevention. While the attack was identified successfully, automated mitigation mechanisms such as firewall rules, intrusion prevention systems (IPS), or rate-limiting controls were not implemented.

12. Conclusion

This project successfully demonstrated SYN flood detection and network geolocation analysis using Wireshark. Packet filtering and GeoIP mapping techniques enabled identification of suspicious network traffic and approximate source location. The controlled simulation of a TCP SYN flood attack using Kali Linux and the `hping3` tool provided practical exposure to the working principles of denial-of-service attacks. By generating millions of TCP SYN packets within a short time frame, the experiment clearly illustrated how attackers can exploit the TCP three-way handshake process to overwhelm a target system. The accumulation of half-open connections demonstrated how server resources such as memory buffers and processing capacity can become exhausted, potentially leading to degraded performance or complete service disruption. Through systematic packet capture and filtering, Wireshark proved to be an effective network monitoring and analysis tool. The application of specific display filters such as `tcp.flags.syn == 1 && tcp.flags.ack == 0` allowed precise isolation of suspicious traffic. Detailed packet inspection at multiple protocol layers (Ethernet II, IPv4, and TCP) enabled extraction of vital information including source IP address, destination IP address, TCP flags, sequence numbers, and MAC addresses. This layered analysis strengthened the understanding of how network attacks are structured at the protocol level. The use of endpoint statistics provided a higher-level analytical view of traffic distribution. By examining packet counts and transmission patterns, it was possible to statistically confirm the abnormal behavior associated with flooding attacks. The large imbalance between transmitted and received packets served as strong evidence of malicious activity. This demonstrates the importance of combining packet-level inspection with statistical analysis for comprehensive network forensics. Furthermore, the integration of the GeoLite2 geolocation database extended the project beyond detection into investigative analysis. Mapping the attacker's IP address to an approximate geographical region provided contextual awareness of the potential origin of the traffic. While the results are limited to ISP-based approximation, this approach illustrates how network traffic data can support preliminary attribution efforts in cybersecurity investigations.

An important outcome of this project is the development of practical skills in packet analysis, attack simulation, and forensic interpretation. Understanding how to capture traffic, apply filters, interpret protocol headers, and analyze communication patterns is essential for cybersecurity professionals. These foundational skills are widely applicable in intrusion detection, incident response, digital forensics, and network security monitoring. Additionally, the project highlights the importance of proactive network monitoring in modern organizations. Early detection of abnormal traffic patterns can prevent large-scale service disruptions and minimize financial or reputational damage. Implementing proper logging, packet analysis tools, and intrusion detection systems can significantly strengthen an organization's security posture. While the experiment was conducted in a controlled local environment, the concepts demonstrated are directly relevant to real-world cybersecurity scenarios. SYN flooding remains a commonly used attack technique, and understanding its detection mechanisms is critical for defending network infrastructure. Future improvements could include implementing automated mitigation techniques such as firewall rate limiting, SYN cookies, intrusion prevention systems (IPS), or machine learning-based anomaly detection.

In conclusion, this project effectively combines theoretical understanding with practical experimentation to demonstrate how SYN flood attacks can be detected and analyzed using Wireshark. The integration of packet capture, statistical analysis, and geolocation mapping provides a comprehensive framework for basic network forensic investigation. As cyber threats continue to grow in complexity and scale, the knowledge and skills developed through this project form a strong foundation for advanced study and professional practice in cybersecurity.

12.1 Future Work

Although the project successfully demonstrated SYN flood detection and IP geolocation analysis in a controlled laboratory environment, several improvements can be implemented in future work.

- **Real-Time Intrusion Detection Integration:** The current implementation focuses on packet capture and post-analysis. Future enhancements can integrate Intrusion Detection Systems (IDS) such as Snort or Suricata for real-time SYN flood detection and alert generation.
- **Automated Alert System:** A notification system can be developed to automatically trigger alerts (email or dashboard notifications) when abnormal SYN packet thresholds are exceeded.
- **Web-Based Monitoring Dashboard:** A centralized dashboard can be created to visualize attack statistics, packet counts, and geolocation results in real time.
- **Advanced Attack Simulation:** Future experiments may include testing additional attacks such as UDP floods, ICMP floods, or distributed SYN flood (DDoS) scenarios.
- **Machine Learning-Based Detection:** Traffic pattern analysis using machine learning models could improve detection accuracy and reduce false positives.
- **Cloud Environment Testing:** The system can be extended to analyze traffic in cloud-based infrastructures for enterprise-level scalability.

These enhancements would improve automation, scalability, and practical applicability of the system in real-world cybersecurity environments.

APPENDIX

A. ENVIRONMENT AND TOOL CONFIGURATION

The experiment was conducted in a controlled laboratory environment to ensure safe and ethical execution of the TCP SYN flood simulation. The systems and tools used during the implementation are detailed below.

System Environment

- Attacker Machine: Kali Linux 2023.4 (64-bit)
- Kernel Version: Linux 6.x
- Victim Machine: Windows 10 Pro (64-bit)
- Packet Analyzer: Wireshark 4.0.x
- GeoIP Database: GeoLite2 City Database (2026 Release)
- Network Configuration: Isolated Private LAN Environment

Tool Version Verification Commands

```
Verify hping3 version hping3 –version
Verify Wireshark version wireshark –version
Verify Linux kernel version uname -r
```

Listing A.1: Environment and Version Verification Commands

B. COMMAND EXECUTION LOG

The following commands were executed during the controlled SYN flood simulation and packet analysis process.

TCP SYN Flood Simulation

```
sudo hping3 -S –flood -p 80 192.168.127.128
```

Wireshark Display Filter

```
tcp.flags.syn == 1  tcp.flags.ack == 0
```

Packet Capture Procedure

1. Launch Wireshark
2. Select active network interface
3. Start packet capture
4. Apply SYN display filter
5. Stop capture
6. Save traffic file as .pcap format

Listing B.1: Complete Command Log Used in the Experiment

C. SYSTEM CONFIGURATION DETAILS

The following configuration steps were applied to ensure accurate packet capture and geolocation mapping.

```
Enable Promiscuous Mode in Wireshark
Capture -> Options -> Enable Promiscuous Mode
Configure GeoIP database
Edit -> Preferences -> Name Resolution -> GeoIP Database Path
Save captured traffic
File -> Save As -> attack_capture.pcap
```

Listing C.1: Configuration and System Reference Commands

D. TOOL SUMMARY AND PURPOSE

- **Kali Linux 2023.4** — Platform used for generating SYN flood traffic.
- **hping3 (3.0.0-alpha-2)** — Packet crafting tool used to simulate TCP SYN flood.
- **Wireshark 4.0.x** — Network protocol analyzer used for packet capture and inspection.
- **GeoLite2 Database** — Used for IP geolocation mapping.

E. Project Reproducibility and File Structure

The following resources are available to ensure independent verification and reproducibility of the experiment:

- PCAP capture file: `attack`$_c$*apture.pcap*
- `Command execution log (documented in Appendix B)`
- `System configuration details (Appendix C)`
- `Tool versions and verification commands (Appendix A)`

All experiments were conducted in a controlled local network environment. The methodology described in Chapters 4–7 allows replication of the experiment using the same tool versions and configuration steps.

For academic submission purposes, external repository links were not included; however, the documented procedure provides complete reproducibility of the setup and analysis process.

References

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